

Практическое занятие 7

Производная

Таблица производных основных элементарных функций

$$(x^\alpha)' = \alpha x^{\alpha-1}, \alpha \neq 0$$

$$(a^x)' = a^x \ln a, a > 0, a \neq 1$$

$$(e^x)' = e^x$$

$$(\log_a x)' = \frac{1}{x \ln a}, a > 0, a \neq 1$$

$$(\ln x)' = \frac{1}{x}$$

$$(\ln|x|)' = \frac{1}{x}$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$$

$$(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$$

$$(\arcsin x)' = -(\arccos x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\operatorname{arctg} x)' = -(\operatorname{arcctg} x)' = \frac{1}{1+x^2}$$

Правила дифференцирования:

$$(C)' = 0$$

$$(f + g)' = f' + g'$$

$$(Cf)' = Cf'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g \neq 0$$

$$(f(g(x)))' = f'(g(x)) g'(x)$$

Частные случаи производной степенной функции:

$$(x)' = 1, \quad (x^2)' = 2x, \quad (x^3)' = 3x^2, \quad (\sqrt{x})' = \frac{1}{2\sqrt{x}}, \quad \left(\frac{1}{x}\right)' = -\frac{1}{x^2},$$

$$\left(\frac{1}{x^2}\right)' = -\frac{2}{x^3}$$

Вычислить производную и ее значение в указанной точке:

$$1) y = 6x^2 - 5x + 1, \quad x_0 = \frac{1}{4}$$

$$y' = 12x - 5, \quad y'\left(\frac{1}{4}\right) = -2$$

$$2) y = 6\sqrt{x} - \frac{5}{\sqrt{x}} + 1, \quad x_0 = \frac{1}{4}$$

$$y' = \frac{3}{\sqrt{x}} + \frac{5}{2\sqrt{x^3}}, \quad y'\left(\frac{1}{4}\right) = 26$$

$$3) y = \frac{2x+1}{3x+2}, \quad x_0 = -1$$

$$y' = \frac{2(3x+2) - 3(2x+1)}{(3x+2)^2} = \frac{1}{(3x+2)^2}, \quad y'(-1) = 1$$

$$4) y = x \ln(x+3), \quad x_0 = -2$$

$$y' = \ln(x+3) + \frac{x}{x+3}, \quad y'(-2) = -2$$

$$5) y = x e^{x^2-1}, \quad x_0 = -1$$

$$y' = e^{x^2-1} + 2x^2 e^{x^2-1} = (1 + 2x^2)e^{x^2-1}$$

$$y'(-1) = 3$$

$$6) y = 2x \sin 3x + 3x \cos 2x, \quad x_0 = \frac{\pi}{2}$$

$$y' = 2 \sin 3x + 6x \cos 3x + 3 \cos 2x - 6x \sin 2x$$

$$y'\left(\frac{\pi}{2}\right) = -5$$

$$7) y = \frac{2\sqrt{x}-3}{4\sqrt{x}-7}, \quad x_0 = 4$$

$$y' = \frac{\frac{1}{\sqrt{x}}(4\sqrt{x}-7) - \frac{2}{\sqrt{x}}(2\sqrt{x}-3)}{(4\sqrt{x}-7)^2} = -\frac{1}{\sqrt{x}(4\sqrt{x}-7)^2}$$

$$y'(4) = -\frac{1}{2}$$

$$8) y = \frac{e^{2x+3}}{e^{3x+2}}, \quad x_0 = 0$$

$$y' = \frac{2e^{2x}(e^{3x} + 2) - 3e^{3x}(e^{2x} + 3)}{(e^{3x} + 2)^2} = \frac{-e^{5x} + 4e^{2x} - 9e^{3x}}{(e^{3x} + 2)^2}$$

$$y'(0) = -\frac{6}{9} = -\frac{2}{3}$$

$$9) y = \arcsin \frac{2x}{5}, \quad x_0 = 2$$

$$y' = \frac{\frac{2}{5}}{\sqrt{1 - \frac{4x^2}{25}}} = \frac{2}{\sqrt{25 - 4x^2}}$$

$$y'(2) = \frac{2}{3}$$

$$10) \quad y = e^{\sin 3x}, \quad x_0 = \frac{\pi}{3}$$

$$y' = 3 \cos 3x e^{\sin 3x}$$

$$y'\left(\frac{\pi}{3}\right) = -3$$

$$11) \quad y = \arccos(\sqrt{x-2}), \quad x_0 = \frac{5}{2}$$

$$y' = -\frac{1}{\sqrt{1-(x-2)}} \frac{1}{2\sqrt{x-2}} = -\frac{1}{2\sqrt{(3-x)(x-2)}}$$

$$y' = \left(\frac{5}{2}\right) = -1$$

$$12) \quad y = 2^{tg 2x}, \quad x_0 = \frac{\pi}{8}$$

$$y' = 2^{tg 2x} \ln 2 \frac{2}{\cos^2 2x} = \frac{2^{tg 2x+1} \ln 2}{\cos^2 2x}$$

$$y'\left(\frac{\pi}{8}\right) = 8 \ln 2$$

$$13) \quad y = \ln \left(\sqrt[7]{\frac{2x-1}{3x+2}} \right), \quad x_0 = 1$$

$$y = \frac{1}{7} (\ln|2x-1| - \ln|3x+2|)$$

$$y' = \frac{1}{7} \left(\frac{2}{2x-1} - \frac{3}{3x+2} \right) = \frac{1}{(2x-1)(3x+2)}$$

$$y'(1) = \frac{1}{5}$$

$$14) \quad y = \ln(4x + \sqrt{16x^2 + 9}), \quad x_0 = 1$$

$$\begin{aligned} y' &= \frac{1}{4x + \sqrt{16x^2 + 9}} \left(4 + \frac{16x}{\sqrt{16x^2 + 9}} \right) \\ &= \frac{1}{4x + \sqrt{16x^2 + 9}} \frac{4(\sqrt{16x^2 + 9} + 4x)}{\sqrt{16x^2 + 9}} = \frac{4}{\sqrt{16x^2 + 9}} \end{aligned}$$

$$y'(1) = \frac{4}{5}$$

$$15) \quad y = \operatorname{arctg} \frac{x+1}{x-1}, \quad x_0 = 3$$

$$y' = \frac{1}{1 + \left(\frac{x+1}{x-1}\right)^2} \left(-\frac{2}{(x-1)^2} \right) = -\frac{2}{(x-1)^2 + (x+1)^2} = -\frac{1}{x^2 + 1}$$

$$y'(3) = -\frac{1}{10}$$

$$16) \quad y = \frac{2 + \sin 3x}{2 - \sin 3x}, \quad x_0 = \frac{\pi}{3}$$

$$y' = \frac{3 \cos 3x (2 - \sin 3x) + 3 \cos 3x (2 + \sin 3x)}{(2 - \sin 3x)^2} = \frac{12 \cos 3x}{(2 - \sin 3x)^2}$$

$$y'\left(\frac{\pi}{3}\right) = -\frac{12}{4} = -3$$

$$17) \quad y = \frac{\ln x}{x}, \quad x_0 = 1$$

$$y' = \frac{\frac{1}{x} x - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$y'(1) = 1$$

$$18) \quad y = \ln(\ln(\frac{1}{x})), \quad x_0 = \frac{1}{e}$$

$$y' = (\ln(-\ln x))' = \frac{1}{x \ln x}$$

$$y'\left(\frac{1}{e}\right) = -e$$

$$19) \quad y = \operatorname{arctg}(\ln x), \quad x_0 = e$$

$$y' = \frac{1}{x(1 + \ln^2 x)}$$

$$y'(e) = \frac{1}{2e}$$

$$20) \quad y = \ln(\operatorname{arctg} x), \quad x_0 = 1$$

$$y' = \frac{1}{(1 + x^2)\operatorname{arctg} x}$$

$$y'(1) = \frac{2}{\pi}$$

$$21) \quad y = \log_{\sin x}(\cos x), \quad x_0 = \frac{\pi}{4}$$

$$\begin{aligned} y' &= \left(\frac{\ln(\cos x)}{\ln(\sin x)} \right)' = \frac{-\frac{\sin x}{\cos x} \ln(\sin x) - \frac{\cos x}{\sin x} \ln(\cos x)}{\ln^2 \sin x} \\ &= \frac{-\sin^2 x \ln(\sin x) - \cos^2 x \ln(\cos x)}{\sin x \cos x \ln^2(\sin x)} \end{aligned}$$

$$y'\left(\frac{\pi}{4}\right) = \frac{-2 \ln \frac{\sqrt{2}}{2}}{\ln^2 \frac{\sqrt{2}}{2}} = \frac{4}{\ln 2}$$